KU LEUVEN

FACULTY OF ENGINEERING TECHNOLOGY

GHENT CAMPUS

Miniature and customer friend solution for tracking and locating insects

Lars DEPREZ

Supervisor(s): Prof. dr. ing. G Callebaut
ing. J Van Moer

Co-supervisor(s): dr. ing. J Van Mulders

Master's Thesis submitted to obtain the degree of Master of Science in ghenteng specialisationprogramme Engineering Technology

Academic year 2026 - 2027

©Copyright KU Leuven

This master's thesis is an examination document that has not been corrected for any errors.

Without written permission of the supervisor(s) and the author(s) it is forbidden to reproduce or adapt in any form or by any means any part of this publication. Requests for obtaining the right to reproduce or utilise parts of this publication should be addressed to KU Leuven, Ghent Campus, Gebroeders De Smetstraat 1, B-9000 Ghent, +32 92 65 86 10 or via email fet.ghent@kuleuven.be.

A written permission of the supervisor(s) is also required to use the methods, products, schematics and programs described in this work for industrial or commercial use, for referring to this work in publications, and for submitting this publication in scientific contests.

Preface

In the preface, you write a personal account with an acknowledgement or expression of gratitude to the people who assisted you in completing your master's thesis and supported you during your studies.



Copyright Information:
student paper as part of an academic education and examination.
No correction was made to the paper after examination.

Samenvatting

Write a short summary in Dutch that is accessible to a wide audience and contains a **maximum of 3500 characters**. This summary must also be uploaded to KU Loket.

Summary

In recent years, the invasive Asian hornet (*Vespa velutina*) has appeared across Europe, posing a significant threat to local wildlife and apiaries. Multiple methods for containing this threat have been invented: including visual tracking, triangulation, harmonic radar and radio telemetry.

This thesis investigates radio telemetry to track Asian hornets. The primary focus is to design a cost-effective, rechargeable, sub-200 mg transmitter that is practical and user-friendly for non-technical operators. The user must be able to attach and activate the transmitter in a simple manner and track the hornet with a portable receiver. This thesis aims to improve upon existing wildlife trackers by being purpose-built specifically for the tracking and eradication of Asian hornets.

The transmitter must minimize energy consumption and component count. This is achieved using a simple, single-stage transmitter designed around a Pierce oscillator combined with a pulsing circuit to conserve energy.

Weight minimization is a critical aspect of this research, offering key practical advantages: reduced tag weight allows a larger battery to be integrated to extend operational lifespan or increase RF output power, while enabling a higher proportion of hornets to carry the payload over greater distances.

The primary goals of this thesis are first to design, simulate, and fabricate an initial transmitter tag, which will be evaluated and refined into an optimized second design iteration. Once a functional tag is realized, a complementary receiver will be designed and tested, leading to field deployment once the tag is small and light enough.

Keywords: Vespa velutina, radio telemetry, Pierce oscillator

Contents

1 introduction **1**

1.1 Asian hornet . . . 1

1.2 Thesis objective . . . 1

1.3 Primary objectives . . . 1

1.3.1 Literature review . . . 1

1.3.2 Theoretical model and simulations . . . 1

1.3.3 Develop initial prototype . . . 1

1.3.4 Optimize the transmitter for hornets . . . 1

1.3.5 Lab testing . . . 1

1.3.6 Field testing . . . 1

2 Literature review **1**

2.1 Introduction . . . 1

2.2 Asian hornet . . . 1

2.3 Existing Hornet-Tracking Methods . . . 1

2.3.1 Visual Tracking . . . 1

2.3.2 Radio Tracking . . . 2

2.3.3 Method Comparison and Selection . . . 2

2.4 Radio-telemetry . . . 2

2.5 Commercial options . . . 2

2.6 Energy storage . . . 3

2.7 Conclusion . . . 3

3 Design and validation **3**

3.1 Component selection and calculation . . . 4

3.1.1 Available parameters . . . 4

3.1.2 Transconductance Calculation . . . 4

3.2 Simulation 5
3.3 Weight analysis 5
3.4 prototype testing 5
3.5 Cost analysis and part availability 5

A The crystal oscillator **7**
A.1 Quartz Crystal 8
A.2 Pierce oscillator 9
A.3 Transconductance calculation 11
A.4 Collector arm 14
A.5 The antenna 15
A.6 Design formula summary 16

B Scientific article describing this thesis **17**

C Poster **19**

D GenAI code of conduct **20**

Acronyms

BJT bipolar junction transistor. i, 3, 11

BLE Bluetooth Low Energy. i, 2

RSSI received signal strength indicator. i, 2

SHF Super High Frequency. i, 2

VHF Very High Frequency. i, 2, 3

Miniature and customer friendly solution for tracking and locating insects

Lars Deprez

1 introduction

1.1 Asian hornet

The Asian hornet forms a problem for both wildlife and human health. Therefore, a cost-effective and easy-to-use tracker is needed to find and remove nests that are a risk to apiaries or human health. Hornet tracking can be split up into two main categories: visual tracking and radio tracking. In visual tracking, the nest is found by visually following the hornet or writing down the flight direction, while radio tracking relies on catching a hornet and applying a tag that can be passive and work with reflection, or that is active and sends out its own radio waves. This thesis focuses on making a radio telemetry tag; this is the active kind.

1.2 Thesis objective

The objective of this thesis is to design a cost-effective, rechargeable, sub-200 mg transmitter that is practical and user-friendly for non-technical operators. The main challenge will be to make a miniature design without losing this usability. This will be achieved by following the primary objectives.

1.3 Primary objectives

1.3.1 Literature review

Here the objective is to learn about the current and past research in the field of radio telemetry and Asian hornets. The commercial trackers will also be compared to look at what the industry is using at the moment. With the gathered information, a design can be chosen that fits this thesis the best.

1.3.2 Theoretical model and simulations

With the design decision made, a theoretical model now needs to be built. This serves the function of understanding how the transmitter works and how the components need to be calculated. The simulation will show if the design and component calculations are correct.

1.3.3 Develop initial prototype

This prototype will validate the simulations and can be experimented with. This prototype will not be miniature.

1.3.4 Optimize the transmitter for hornets

The objective of this part will be to scale down the transmitter to sub-200 mg and make the tag attachable to hornets.

1.3.5 Lab testing

This part aims to validate the range, frequency stability, battery life, and fine-tune where needed.

1.3.6 Field testing

At this point, the tag is ready to be tested in the field. The main objective is to validate the hornet attachment mechanism and to see how well hornets cope with flying with the tag.

2 Literature review

2.1 Introduction

The literature review looks at the impact Asian hornets have on their environment and different methods of tracking Asian hornets with their advantages and disadvantages. Further, we investigate radio telemetry in more detail, look at commercial tag and battery options and end with a conclusion.

2.2 Asian hornet

According to Lioy et al. [1], the Asian hornet was introduced into northern France by boat in 2004 and subsequently spread throughout northern Europe. The authors note that this invasive species preys on 159 different insects, including honey bees, consuming up to 11.32 kg of insect biomass per season while actively competing with native hornets. Beyond the ecological damage, the invasion carries significant socio-economic consequences. Hornets defending their colonies pose a risk of severe injury or death to humans, and the financial burden of protecting bee populations and managing the invasion can reach 29.5 million euros annually.

The question of how to manage or even if there should be management will not be answered because it is an ongoing debate; however, this thesis will look into how to find hornet nests in a cost-effective and user-friendly way.

2.3 Existing Hornet-Tracking Methods

The ways of tracking hornets can be split into two groups: visual tracking and radio tracking. They each have their own advantages and disadvantages.

2.3.1 Visual Tracking

Currently, three visual tracking options are commonly used. The first is attaching a lightweight identifier, like reflective tape, to the hornet's waist; this is a cost-effective solution that improves upon tracking with the naked eye, though operators can still easily lose line of sight when the hornet flies

out of range [2]. The second option is triangulation, which works by releasing captured hornets from different locations and plotting their flight directions on a map. If the hornets come from the same colony, their flight paths should point to the nest. While this method is intuitive and easy to learn, it is time-consuming, and hornets do not always fly in a straight line to their nest, which can make flight directions inaccurate [2]. The third option is thermal imaging. This works because Asian hornet nests actively regulate their own temperature. A nest becomes visible on thermal images when there is a temperature difference compared to the surrounding environment. This method is most effective in the early morning, without dense tree canopy and when the operator is close to the nest [3].

2.3.2 Radio Tracking

The first technique is harmonic radar, which utilizes a lightweight transponder (20 mg) featuring a passive diode that reflects the signal back to the receiver. Operating within the Super High Frequency (SHF) band, these systems allow for miniature tag dimensions. Harmonic radar determines a target's position by processing the backscattered signal, where time delay indicates range and the angle of arrival provides the direction. Tracking the target across consecutive snapshots reveals both flight trajectory and velocity. The primary limitation of harmonic radar is its high susceptibility to clutter from nearby obstacles, such as dense vegetation or artificial structures, generally restricting reliable operation to open environments [2].

The final approach is radio telemetry. This method involves mounting a miniature active radio transmitter (less than 200 mg) onto the hornet. The direction is determined using a handheld Yagi antenna, while distance can be found from the received signal strength indicator (RSSI). The main constraint of radio telemetry lies in the significant design effort required to miniaturize the circuitry and power consumption. Nevertheless, its primary advantage is active signal transmission, which substantially simplifies signal detection and homing [2].

2.3.3 Method Comparison and Selection

Combining these distinct tracking methodologies can enhance overall search success rates. For instance, active radio telemetry can be deployed for long-range directional tracking, while thermal imaging can be utilized upon reaching the general vicinity to locate the thermal signature of the hidden nest. Consequently, this thesis focuses on the development of an active radio telemetry transmitter to leverage continuous signal broadcasting. This avoids the clutter limitations of harmonic radar in complex terrain, which is essential given that hornet nests are frequently located in woodland and urban environments [1].

The central engineering challenge lies in balancing competing design requirements: minimizing physical mass and payload volume while maximizing power efficiency and radiated RF output within a single optimized architecture.

2.4 Radio-telemetry

The field of radio telemetry has seen significant evolution since the first radio tag was introduced in 1963 [4]. This tag (Figure 2.1) used a single transistor, crystal, capacitor, resistor, loop antenna, and battery to form a transmitter. The loop antenna was used as a collar around the animal being tracked, and the complete tag weighed 10 g.

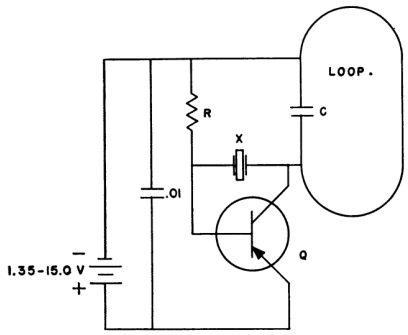


Figure 2.1: Radio-Tracking system for wild animals [4]

This type of transmitter is often called a single-stage tag and operates in the Very High Frequency (VHF) band. Cochran's original design transmitted a signal continuously, which is usually unnecessary. Therefore, it is common to modify the circuit with an extra RC network at the base of the transistor to create a pulsing circuit [5].

While the single-stage tag is simple and requires few components, it has a key flaw: changes in load impedance directly affect the oscillation frequency. This change in impedance occurs when the antenna moves closer to or farther from the animal. Consequently, two-stage tags were developed to improve frequency stability. This extra complexity adds weight to the tag while being a limited improvement for hornet tags.

In recent years, Bluetooth Low Energy (BLE) tags have emerged. They utilize the Bluetooth protocol and can be programmed to distinguish between individual tags. These tags operate at 2.4 GHz, resulting in a much smaller antenna compared to single-stage and two-stage transmitters, but higher frequencies have a limited range and suffer more attenuation from vegetation.

This means that the single-stage tag remains the favorite for small insect trackers due to its low complexity and the ability to use a lower frequency compared to BLE.

2.5 Commercial options

An important part of the literature study is to find commercial tags and compare them to each other, learn about different technologies that are used in

practice, and get an understanding of acceptable prices, ranges, tag weights, and tag dimensions. Different commercial tags are shown in Table 2.1, some purpose-built for tracking hornets while others have more general use cases. The system cost is the price of all the equipment needed to start tracking. The VESPA tracker has the lowest price per tag but lacks in range. The VFT 110 has the lightest tag with a good range and tag price, but the system cost is higher than that of the VESPA and Intuite trackers.

2.6 Energy storage

The majority of the tag weight will be in the energy storage; this needs to suffice for more than 2 hours of operation. The main battery requirements are a sub-100 mg weight, adequate capacity between 0.1 mAh and 1 mAh, and a small footprint within 5×5 mm.

Table 2.2 shows different battery options with relevant specs. The two options are LiPo batteries with higher capacity and solid-state ceramic batteries with higher discharge current. The low discharge current of the LiPo battery can be compensated with a capacitor as long as the average current does not exceed the discharge current and the capacity is sufficiently large. Charging the LiPo battery is more complex and time-consuming, requiring a current- and voltage-controlled charger, while solid-state ceramic batteries can be fully charged within minutes using just a fixed voltage source.

2.7 Conclusion

The literature review shows that Asian hornets pose an ecological, economic, and health threat, and an effective method of tracking their nests is essential to protect beehives and human health. While visual tracking and harmonic radar are limited by human eyesight and environmental clutter, the single-stage VHF transmitter emerges as the optimal choice. It provides active signal broadcasting while remaining lightweight (<200 mg) with a minimal component count.

Building upon these findings, Chapter 3 details the circuit design, component selection, and SPICE simulations for the VHF single-stage transmitter prototype.

3 Design and validation

The design of a single-stage transmitter is built around a Pierce oscillator (shown in Figure 3.1); this is a circuit built out of a crystal, two capacitances, and an inverter. It works by inducing a 180° phase shift through the crystal and capacitance network and then inverting and amplifying that signal. This circuit has a total 360° phase shift and an amplification greater than 1, fulfilling the Barkhausen criterion. The single-stage transmitter makes use of a crystal, and the two capacitances come from the base-emitter capacitance of the transistor and the capacitively tuned LC tank network. The transistor acts as an inverter/amplifier. The base resistor regulates the bias current, and the base RC network makes the oscillator pulse. A detailed explanation of the oscillator circuit can be found in Appendix A.

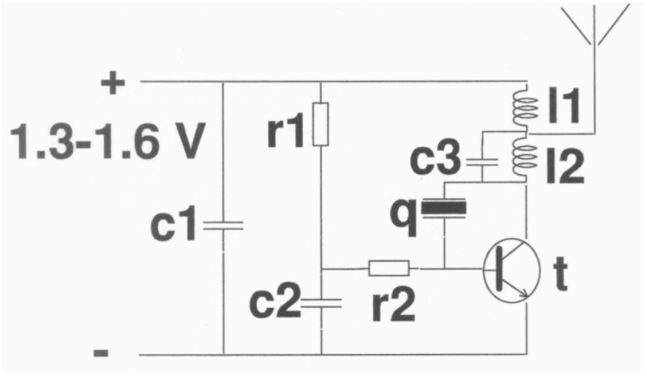


Figure 3.1: Schematic of the single-stage Pierce oscillator circuit (taken from [8]).

The circuit has multiple requirements to be able to oscillate. The collector side branch needs to be capacitively tuned for the oscillation frequency of the crystal. This means that the LC tank frequency must be lower than the oscillation frequency, which is capped by an upper boundary formed by L_1 and the antenna. This relationship is defined in Equation 3.1, where L_1' is the equivalent inductance resulting from L_1 and the antenna. This is detailed in Appendix A:

$$\omega_{\text{tank}} < \omega_{\text{osc}} < \omega_{\text{tank}} \sqrt{1 + \frac{L_2}{L_1'}} \tag{3.1}$$

In an ideal case, L_1 will be tuned to form a resonant circuit with the antenna at the oscillation frequency, but this is not always possible.

The second requirement states that the base-emitter of the transistor needs to be capacitive in the pF order of magnitude. Some base-emitter capacitances of bipolar junction transistors (BJTs) are too low, especially from RF BJTs, requiring a capacitor to be placed between the pins.

The transconductance g_m is fundamentally dependent on the DC collector current. By substituting $I_c = \beta I_b$, the transconductance can be expressed directly as a function of the base bias current:

$$g_m = \frac{I_c}{V_T} = \frac{\beta I_b}{V_T} \tag{3.2}$$

The transistor's transconductance generates an active negative resistance that amplifies the signal to overcome the parasitic losses within the oscillation circuit, primarily the crystal's motional resistance (R_s), and the losses caused by the capacitors. The minimum critical transconductance required for oscillation startup is expressed as:

$$g_{m,\text{crit}} = 4\omega^2 R_s C_T^2 \tag{3.3}$$

Table 2.1 Comparison of existing tracking tags suitable for Asian hornets as of October 2026.

Tracker	Tag Cost (€)	System Cost (€)	Range (m)	Battery (h)	Weight (mg)
VESPA tracker	21.6	620	500	4	125
Intuite	63	348	4000	4	114
Robor VFT 110	49	1795	2000	144	110
Lotek (Nanotag) NTP-1	190	3215	1500	288	130

Table 2.2 Comparison of selected Seiko and ITEN micro-batteries highlighting technology, capacity, weight, and discharge current [6] [7].

Model	Technology	Capacity (mAh)	Weight (mg)	Discharge Current (μA)
Seiko MS412FE	LiPo	1	70	10
Seiko MS414GE	LiPo	2	80	10
ITEN PWY0150S	Solid state ceramic	0.15	73	1000
ITEN PWY0250S	Solid state ceramic	0.25	95	1000

A detailed derivation and full parameter analysis are given in Appendix A.

3.1 Component selection and calculation

This section shows how the calculations and component selection are done in the thesis. First, the crystal and transistor need to be selected. The crystal will determine the operation frequency, which is the third harmonic in this thesis. It is important to choose a crystal with low load capacitance and motional resistance to keep $g_{m,\text{crit}}$ as low as possible. The most important parameter of the transistor is the f_T frequency, which should be 5x to 10x above the operating frequency. Keeping the C_{cb} capacitance as low as possible will also keep $g_{m,\text{crit}}$ to a minimum. All other components are passive and have fewer requirements; only the capacitors and inductors on the collector side still have to function as capacitors and inductors at the operating frequency.

3.1.1 Available parameters

The following list shows the selected components and the variables given by their specifications:

- **Crystal:** Gives the oscillation frequency (f_{osc}), static capacitance (C_0), and motional resistance (R_s).
- **Transistor:** Gives the feedback capacitance (C_μ) and base-emitter capacitance (C_π).

Additionally, for the circuit calculations, we assume that the total base capacitance (C_b) and the total collector capacitance (C_c) are approximately the same ($C_b \approx C_c$).

With these variables, the external load capacitance (C_L) and the total effective parallel capacitance (C_T) can be calculated using the following formulas:

$$C_L = \frac{C_b C_c}{C_b + C_c} \tag{3.4}$$

$$C_T = C_0 + C_L + C_\mu \tag{3.5}$$

3.1.2 Transconductance Calculation

The critical transconductance ($g_{m,\text{crit}}$) required for oscillation can be calculated using the parameters established in the previous section (where $\omega = 2\pi f_{\text{osc}}$):

$$g_{m,\text{crit}} = \omega^2 C_b C_c R_s \left(\frac{C_T}{C_L}\right)^2 \tag{3.6}$$

To ensure a reliable oscillator startup under various conditions, the target transconductance (g_m) is designed to be five times higher than this critical threshold:

$$g_m = 5 \times g_{m,\text{crit}} \tag{3.7}$$

Using this target transconductance, the required DC base bias current (I_b) and the values for the bias network resistors (R_1 and R_2) can be determined. The base current is derived using the thermal voltage (V_T) and the transistor's common-emitter current gain (β):

$$I_b = \frac{g_m V_T}{\beta} \tag{3.8}$$

Assuming a standard series biasing configuration from the supply voltage (V_{CC}), the total required bias resistance is determined by Ohm's law. Consequently, the sum of the bias resistors R_1 and R_2 must equal:

$$R_1 + R_2 = \frac{V_{CC} - V_{BE}}{I_b} \tag{3.9}$$

3.1.3 collector arm calculation

The main requirement of the collector arm is that the total impedance needs to be capacitive for the oscil-

laton to start as stated in Appendix A. The values for L1 L2 and C3 need to be calcualted and the only unknow is Ca, this is the capacitance of the antenna and can be measured using a vector network analyzer (VNA)

3.2 Simulation

3.3 Weight analysis

Due to the strict 200 mg weight there is a need to calculate and approximate the transmitter weight before an actual prototype is made. This will be done by analysing all the components separate and adding up their weights. The pcb weight will be calculated via its area, copper area will also be taken into account.

Resistor weight 0.17 mg per part (https://www.vishay.com/docs/20131/crcw0201ate3.pdf)

Capacitor weight 0.33 mg per part (url = `https://example.com/products/capacitor/ceramiccapacitor/weight_table.ashx?la=en&hash=123`)

Inductor weigth 0.28 mg per part (https://docs.rs-online.com/9d9f/0900766b81582057.pdf)

Transistor weight 0.9 mg per part (https://www.mouser.com/datasheet/2/115/MMBT2222ALP4-219422.pdf)

Battery weight 80 mg per part [6]

Crystal weight 3 mg per part (url = `https://download.epsondevice.com/td/pdf/app/FA-118T_en.pdf?_gl=1*phzuqp*_ga*NzY5NTUyNzU3LjE3OTQyMTM1NDE.*_ga_2TQLB6HKRY*czE3OTQyMTM1NDAkbzEkZzEkdDE3OTQyMTM1NzYkajI0JGwwJGgw`)

PCB weight

3.4 prototype testing

3.5 Cost analysis and part availability

Bibliography

[1] S. Lioy, C. Bergamino, and M. Porporato, "The invasive hornet Vespa velutina : Distribution, impacts and management options", *CABI Reviews*, vol. 2022, Oct. 19, 2022. doi: 10.1079/cabireviews202217030

[2] C. Wang, J. Huang, X. Wan, and Z. Guo, "A Review of Five Existing Hornet-Tracking Methods", *Insects*, vol. 15, no. 8, p. 601, Aug. 2024, issn: 2075-4450. doi: 10.3390/insects15080601 Accessed: Oct. 27, 2026. [Online]. Available: https://www.mdpi.com/2075-4450/15/8/601

[3] S. Lioy, E. Bianchi, A. Biglia, M. Bessone, D. Laurina, and M. Porporato, "Viability of thermal imaging in detecting nests of the invasive hornet Vespa velutina", issn: 1744-7917. doi: 10.1111/1744-7917.12760 Accessed: Oct. 27, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1111/1744-7917.12760?casa_token=ieq3pWHHHQQAAAAA%3AWmU8mPBpobGcvBshUQS9U_rJ_gJvSrVSGYAO_EVT-UpwQcVcXer118cxmh1xh2J905ShdrRq6yepsP8J

[4] W. W. Cochran and R. D. Lord, "A Radio-Tracking System for Wild Animals", *The Journal of Wildlife Management*, vol. 27, no. 1, pp. 9–24, 1963, issn: 0022-541X. doi: 10.2307/3797775 JSTOR: 3797775. Accessed: Oct. 27, 2026. [Online]. Available: https://www.jstor.org/stable/3797775

[5] S. Toledo, "A Review of Miniature Radio Transmitters for Wildlife Tracking", *Sensors (Basel, Switzerland)*, vol. 25, no. 2, p. 517, Jan. 17, 2025, issn: 1424-8220. doi: 10.3390/s25020517 PMID: 39860887. Accessed: Oct. 27, 2026. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11768426/

[6] "Micro Battery (MS Rechargeable Battery, Chip Capacitors, Silver Oxide Battery) | Seiko Instruments Inc. Micro Energy Division", Accessed: Nov. 9, 2026. [Online]. Available: https://www.sii.co.jp/en/me/battery/

[7] "Explore ITEN's Range of Solid-State Ceramic (SSC) Devices", Accessed: Nov. 9, 2026. [Online]. Available: https://www.iten.com/products

[8] B. Naef-Daenzer, "A New Transmitter for Small Animals and Enhanced Methods of Home-Range Analysis", *The Journal of Wildlife Management*, vol. 57, no. 4, pp. 680–689, 1993, issn: 0022-541X. doi: 10.2307/3809066 JSTOR: 3809066. Accessed: Oct. 13, 2026. [Online]. Available: https://www.jstor.org/stable/3809066

Appendix A

The crystal oscillator

In this appendix, the theory of the single-stage transmitter is explained. This is a simple transmitter with one transistor and a crystal. The explanation is divided into multiple parts, starting with the theory of the crystal, followed by the Pierce circuit. Then, we calculate the minimum transconductance needed by the transistor, and finally end with the design of the LC tank circuit and antenna.

Figure A.1 displays the single-stage transmitter circuit, which will be analyzed step by step in the following sections.

This circuit works based on two 180° phase shifts of the signal: once by the transistor and once by the crystal. The condition when this phase shift occurs is described below.

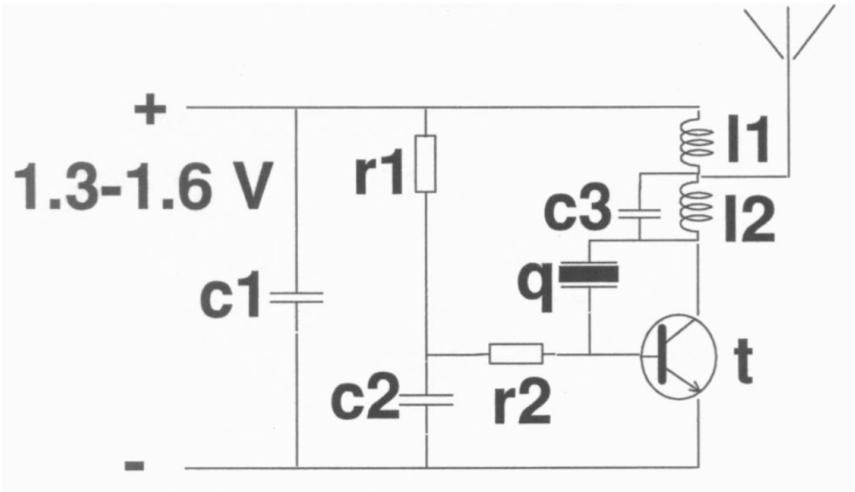


Figure A.1: Schematic of the single-stage Pierce oscillator circuit (taken from [8]).

A.1 Quartz Crystal

The quartz crystal sits centrally in the schematic and creates the stable RF signal of the circuit. To achieve the required VHF operating frequency while keeping the crystal physically robust, the crystal is operated in its 3rd overtone mode (approximately three times its fundamental frequency). To analyze its electrical behavior and frequency response, the crystal is represented by the Butterworth-Van Dyke (BVD) equivalent circuit model shown in Figure A.2.

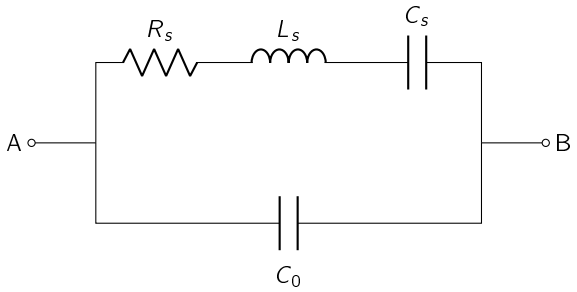


Figure A.2: Butterworth-Van Dyke (BVD) equivalent circuit model of a quartz crystal showing the series motional branch (R_s, L_s, C_s) in parallel with the static shunt capacitance (C_0) (own work).

The frequency response is derived by combining the series and parallel branches together and determining the resonance conditions.

$$Z_s(j\omega) = j\omega L_s + \frac{1}{j\omega C_s} = \frac{1-\omega^2 L_s C_s}{j\omega C_s}, \quad Z_0(j\omega) = \frac{1}{j\omega C_0} \tag{A.1}$$

Combining both equations in parallel yields:

$$\begin{aligned} Z(j\omega) &= Z_s(j\omega) \parallel Z_0(j\omega) \\ &= \frac{\left(\frac{1-\omega^2 L_s C_s}{j\omega C_s}\right) \cdot \left(\frac{1}{j\omega C_0}\right)}{\left(\frac{1-\omega^2 L_s C_s}{j\omega C_s}\right) + \frac{1}{j\omega C_0}} \\ &= \frac{1-\omega^2 L_s C_s}{j\omega\left[(C_s + C_0) - \omega^2 L_s C_s C_0\right]} \\ &= \frac{1}{j\omega C_0} \cdot \frac{\frac{1}{L_s C_s} - \omega^2}{\frac{1}{L_s}\left(\frac{1}{C_s} + \frac{1}{C_0}\right) - \omega^2} \end{aligned} \tag{A.2}$$

Finding the zeros and poles of $Z(j\omega)$ shows the resonant frequencies of the quartz crystal:

1. **Series Resonant Frequency (f_s):** Occurs when $Z(j\omega) = 0$:

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \tag{A.3}$$

2. **Parallel Resonant Frequency (f_p):** Occurs when $Z(j\omega) \to \infty$. Factoring out C_s directly relates f_p to f_s:

$$f_p = \frac{1}{2\pi}\sqrt{\frac{1}{L_s C_s}\left(1 + \frac{C_s}{C_0}\right)} = f_s\sqrt{1 + \frac{C_s}{C_0}} \tag{A.4}$$

Since $C_0 \gg C_s$, the ratio $\frac{C_s}{C_0} \ll 1$. Applying a first-order Taylor series approximation ($(1 + x)^{1/2} \approx 1 + \frac{x}{2}$) provides a simplified expression for the parallel resonant frequency:

$$f_p \approx f_s\left(1 + \frac{C_s}{2C_0}\right) \tag{A.5}$$

For $f < f_s$ and $f > f_p$, the circuit behaves like a capacitor, whereas for $f_s < f < f_p$, it behaves like an inductor. These properties will be important when defining the oscillation conditions in the Pierce circuit.

A.2 Pierce oscillator

The Pierce oscillator works on the basis that a quartz crystal and two capacitors will induce a 180° phase shift onto a signal. Inverting that signal then results in an oscillation following the Barkhausen criteria, meaning the total phase shift must be $k \cdot 360°$ ($k \in \mathbb{N}$) and the amplification should be ≥ 1.

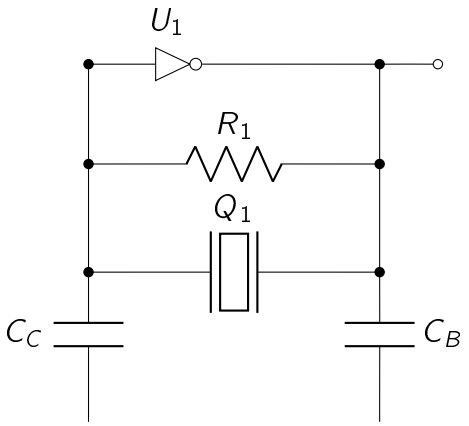


Figure A.3: A Pierce oscillator circuit.

In the following steps, C_b and C_c represent the total node capacitances from base to ground and collector to ground, respectively. The feedback resistor and inverter are removed in this analysis, as the transistor functions as the inverting element.

We start by modeling the collector-to-base voltage ratio using a voltage divider formed by the crystal reactance X_q and the base capacitance C_b:

$$Z_b(j\omega) = \frac{1}{j\omega C_b}, \quad Z_q(j\omega) = jX_q \tag{A.6}$$

Applying the voltage divider rule yields the base voltage V_b relative to the collector voltage V_c:

$$\begin{aligned}\frac{V_b}{V_c} &= \frac{Z_b(j\omega)}{Z_q(j\omega) + Z_b(j\omega)} \\ &= \frac{\frac{1}{j\omega C_b}}{jX_q + \frac{1}{j\omega C_b}} \\ &= \frac{1}{1 - \omega C_b X_q}\end{aligned} \tag{A.7}$$

To achieve the required 180° phase shift, the transfer ratio must be negative ($\frac{V_b}{V_c} < 0$). This requires the denominator $1 - \omega C_b X_q$ to be negative. Since ω and C_b are strictly positive, X_q must be positive ($X_q > 0$), confirming that the crystal must operate in its inductive reactance region ($f_s < f < f_p$).

An important effect of the node capacitances C_b and C_c is that they alter the resonant frequency of the quartz crystal, which must be taken into account. Looking from the crystal terminals, C_b and C_c are in series with each other and in parallel with C_0. The total external load capacitance C_L experienced by the crystal is defined as:

$$C_L = \frac{C_b C_c}{C_b + C_c} \tag{A.8}$$

In addition to the static capacitance C_0 and the external load capacitance C_L, the internal transistor feedback capacitance C_μ (between collector and base) also sits directly in parallel across the crystal terminals. Combining these contributions yields the total effective parallel capacitance C_T:

$$C_T = C_0 + C_L + C_\mu = C_0 + \frac{C_b C_c}{C_b + C_c} + C_\mu \tag{A.9}$$

Consequently, the loaded parallel resonant frequency f_p expressed in terms of C_T becomes:

$$f_p = f_s \sqrt{1 + \frac{C_s}{C_T}} \tag{A.10}$$

Applying the first-order Taylor approximation ($(1 + x)^{1/2} \approx 1 + \frac{x}{2}$) yields the simplified expression for the loaded frequency:

$$f_p \approx f_s \left(1 + \frac{C_s}{2C_T}\right) \tag{A.11}$$

A.3 Transconductance calculation

In this section, the critical transconductance ($g_{m,\text{crit}}$) of the transistor is derived to determine the threshold condition for oscillation startup. This is done by deriving the negative resistance generated by the BJT and adjusting for losses inside the quartz crystal.

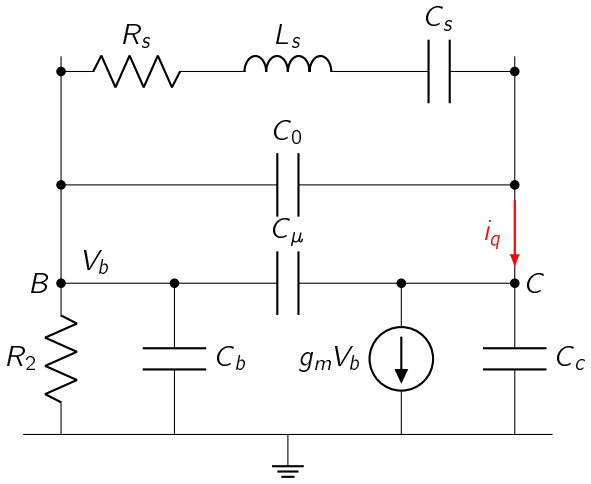


Figure A.4: Small-signal equivalent circuit of the BJT Pierce oscillator with current i_q flowing from the crystal feedback network into the collector node.

Node B Analysis

The bias resistor R_2 can be neglected assuming it is sufficiently large.

$$V_b = -\frac{i_q}{j\omega C_b} \tag{A.12}$$

Node C Analysis (KCL)

Applying KCL at Node C, the sum of currents entering the node equals the sum of currents leaving to ground.

$$i_q = g_m V_b + j\omega C_c V_c \tag{A.13}$$

Solving for V_c directly from the Node C:

$$V_c = \frac{i_q - g_m V_b}{j\omega C_c} \tag{A.14}$$

Substituting V_b into this expression gives:

$$\begin{aligned} V_c &= \frac{i_q - g_m\left(-\frac{i_q}{j\omega C_b}\right)}{j\omega C_c} \\ &= \frac{i_q}{j\omega C_c} - \frac{g_m i_q}{\omega^2 C_b C_c} \end{aligned} \tag{A.15}$$

Combining these two nodes with i_q results into the impedance seen by the quartz crystal.

$$\begin{aligned} V_c - V_b &= \left(\frac{i_q}{j\omega C_c} - \frac{g_m i_q}{\omega^2 C_b C_c}\right) - \left(-\frac{i_q}{j\omega C_b}\right) \\ &= i_q\left(\frac{1}{j\omega C_b} + \frac{1}{j\omega C_c} - \frac{g_m}{\omega^2 C_b C_c}\right) \\ &= i_q\left(-\frac{g_m}{\omega^2 C_b C_c} - j\frac{C_b + C_c}{\omega C_b C_c}\right) \end{aligned} \tag{A.16}$$

Dividing by i_q defines the active transistor input impedance Z_t seen by the crystal:

$$Z_t = \frac{V_c - V_b}{i_q} = -\frac{g_m}{\omega^2 C_b C_c} - \frac{j}{\omega C_L} \tag{A.17}$$

where $C_L = \frac{C_b C_c}{C_b + C_c}$ is the equivalent series load capacitance. The negative real part $R_a = -\frac{g_m}{\omega^2 C_b C_c}$ provides the active power necessary to overcome system losses and initiate oscillation.

Power Balance and Critical Transconductance

To determine the startup threshold condition, the crystal current i is modeled as dividing between two parallel branches across the terminal voltage v:

- **Branch A (Active Circuit):** Impedance $Z_t = R_a - \frac{j}{\omega C_L}$
- **Branch P (Parasitics):** Combined capacitance $C_p = C_0 + C_\mu$, with impedance $Z_p = \frac{1}{j\omega C_p}$

Applying Kirchhoff's Current Law (KCL) at the node and expressing the currents using branch admittances ($Y = \frac{1}{Z}$):

$$i = i_a + i_p \tag{A.18}$$

$$v = i_a Z_t = i_p Z_p \implies i_a = vY_t, \quad i_p = vY_p \tag{A.19}$$

Summing the branch currents allows v to be expressed in terms of the total current i:

$$i = v(Y_t + Y_p) \implies v = \frac{i}{Y_t + Y_p} \tag{A.20}$$

Substituting v back into the active branch current $i_a = vY_t$ gives the exact current divider equation:

$$i_a = i\frac{Y_t}{Y_t + Y_p} \tag{A.21}$$

The branch admittances are defined as $Y_p = j\omega C_p$ and $Y_t = \frac{1}{R_a - j/(\omega C_L)}$. Assuming $|R_a| \ll \frac{1}{\omega C_L}$, the active impedance is dominated by its capacitive term ($Z_t \approx -\frac{j}{\omega C_L}$), simplifying the active admittance to $Y_t \approx j\omega C_L$.

Substituting Y_t and Y_p back into the current divider yields:

$$\begin{aligned} i_a &= i\frac{j\omega C_L}{j\omega C_L + j\omega C_p} \\ &= i\frac{C_L}{C_L + C_p} \\ &= i\frac{C_L}{C_L + C_0 + C_\mu} = i\frac{C_L}{C_T} \end{aligned} \tag{A.22}$$

where $C_T = C_L + C_0 + C_\mu$ represents the total parallel capacitance.

Real power is exchanged solely within the active resistance R_a (driven by i_a) and the crystal's series loss resistance R_s (driven by i):

$$P_{\text{supplied}} = |R_a| i_a^2 = |R_a| \left(\frac{C_L}{C_T}\right)^2 i^2 \tag{A.23}$$

$$P_{\text{loss}} = R_s i^2 \tag{A.24}$$

Equating $P_{\text{supplied}} = P_{\text{loss}}$ and canceling i^2:

$$R_s = |R_a| \left(\frac{C_L}{C_T}\right)^2 \tag{A.25}$$

Substituting $|R_a| = \frac{g_m}{\omega^2 C_b C_c}$ yields the critical transconductance $g_{m,\text{crit}}$ required for oscillation startup:

$$g_{m,\text{crit}} = \omega^2 C_b C_c R_s \left(\frac{C_T}{C_L}\right)^2 \tag{A.26}$$

Substituting $C_L = \frac{C_b C_c}{C_b + C_c}$ into this expression simplifies the relationship by canceling common $C_b C_c$ terms:

$$g_{m,\text{crit}} = \omega^2 R_s C_T^2 \frac{(C_b + C_c)^2}{C_b C_c} = \omega^2 R_s C_T^2 \left(\frac{C_b}{C_c} + 2 + \frac{C_c}{C_b} \right) \tag{A.27}$$

This form shows that $g_{m,\text{crit}}$ depends primarily on the total load capacitance C_T^2 and the capacitance ratio asymmetry, reaching its minimum value of $g_{m,\text{crit, min}} = 4\omega^2 R_s C_T^2$ when $C_b = C_c$.

The resulting $g_{m,\text{crit}}$ is used to calculate the required base bias current. To ensure a reliable startup, it is recommended to maintain a design margin of $g_m = (3 \text{ to } 5) \times g_{m,\text{crit}}$.

The DC collector bias current I_c is determined from the transconductance g_m and the thermal voltage V_T:

$$I_c = g_m V_T \tag{A.28}$$

where $V_T = \frac{k_B T}{q} \approx 25.85$ mV at room temperature (300 K).

The corresponding base bias current I_b is obtained using the transistor's common-emitter current gain β:

$$I_b = \frac{I_c}{\beta} = \frac{g_m V_T}{\beta} \tag{A.29}$$

Keep in mind that V_T and β fluctuate with temperature and operating conditions, so extreme values should be taken into account when creating a practical design.

A.4 Collector arm

The Pierce oscillator requires a capacitive reactance on both sides of the quartz crystal. On the collector side, this capacitance is named C_c, but in reality, it consists of a network comprising two inductors, a capacitor, and an antenna. For the oscillator to operate, this network must have a net capacitive reactance at the oscillation frequency.

The lower bound of the capacitive frequency range for the collector arm is defined by $\omega_{\text{tank}} = \frac{1}{\sqrt{L_2 C_3}}$, where $L_1 \parallel C_a$ is simplified to $L_1' = \frac{L_1}{1-\omega^2 L_1 C_a}$.

The upper bound is obtained by requiring a net capacitive reactance:

$$\omega L_1' + \frac{\omega L_2}{1 - \omega^2 L_2 C_3} < 0 \tag{A.30}$$

Dividing by $\omega > 0$ and multiplying by $(1 - \omega^2 L_2 C_3) < 0$ (which reverses the inequality sign since $\omega > \omega_{\text{tank}}$) yields:

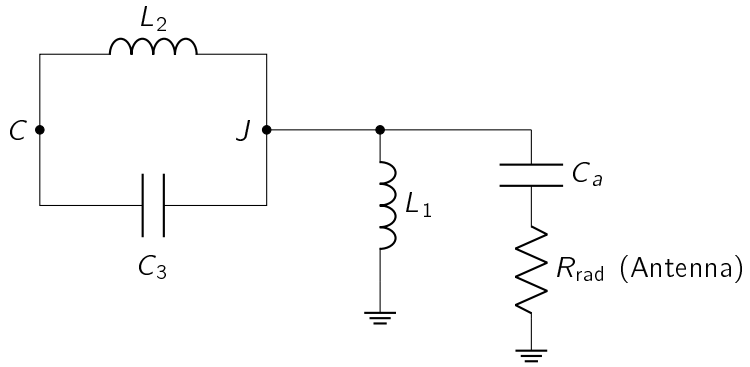


Figure A.5: Equivalent circuit representation of the collector network connected to the parallel $L_2//C_3$ tank, L_1 matching inductor, and the antenna load model (C_a, R_{rad}).

$$\begin{aligned} L_2 &> -L_1'\left(1-\omega^2 L_2 C_3\right) \\ L_1' + L_2 &> \omega^2 L_1' L_2 C_3 \\ \omega^2 &< \frac{1}{L_2 C_3}\left(1+\frac{L_2}{L_1'}\right) \end{aligned} \tag{A.31}$$

Substituting $\omega_{\text{tank}}^2 = \frac{1}{L_2 C_3}$ defines the valid frequency range for capacitive collector behavior:

$$\omega_{\text{tank}} < \omega_{\text{osc}} < \omega_{\text{tank}}\sqrt{1+\frac{L_2}{L_1'}} \tag{A.32}$$

The tank circuit must be designed such that the oscillation frequency falls within these frequency bounds.

A.5 The antenna

The transmitter will use a monopole antenna; these have the main advantage of a small footprint, which is crucial. The main drawback is that a monopole requires a ground plane; without this, the efficiency drops.

The monopole needs to be ideally $\lambda/4$ long, this is the standard quarter-wave monopole. The physical length l of a quarter-wave antenna can be calculated using the speed of light c and the operating frequency f:

$$l = \frac{\lambda}{4} = \frac{c}{4f} \tag{A.33}$$

This is still too long for lower frequencies. To show the impact of the frequency on the antenna size: at 150 MHz ($\lambda = 2.0\,\text{m}$), a quarter-wave antenna is 50 cm long. At 440 MHz ($\lambda = 68.2\,\text{cm}$), it is about 17 cm long. Only at higher frequencies like 2.4 GHz ($\lambda = 12.5\,\text{cm}$) does the antenna become highly compact at roughly 3.1 cm.

A.6 Design formula summary

For convenience during the practical design phase, all critical equations required to calculate the components and operating conditions of the single-stage Pierce oscillator are summarized below.

Crystal and Load Capacitance

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \quad \text{(Intrinsic series resonant frequency)} \tag{A.34}$$

$$C_L = \frac{C_b C_c}{C_b + C_c} \quad \text{(External load capacitance)} \tag{A.35}$$

$$C_T = C_0 + C_L + C_\mu \quad \text{(Total effective parallel capacitance)} \tag{A.36}$$

$$f_L \approx f_s \left(1 + \frac{C_s}{2C_T}\right) \quad \text{(Loaded operating frequency)} \tag{A.37}$$

Transistor Startup and Biasing

$$g_{m,\text{crit}} = \omega^2 R_s C_T^2 \left(\frac{C_b}{C_c} + 2 + \frac{C_c}{C_b}\right) \quad \text{(Minimum critical transconductance)} \tag{A.38}$$

$$g_m = (3 \text{ to } 5) \times g_{m,\text{crit}} \quad \text{(Target transconductance with safety margin)} \tag{A.39}$$

$$I_c = g_m V_T \quad \text{(Required DC collector bias current)} \tag{A.40}$$

$$I_b = \frac{I_c}{\beta} \quad \text{(Required DC base bias current)} \tag{A.41}$$

Collector Network and Antenna Matching

$$L_1' = \frac{L_1}{1 - \omega^2 L_1 C_a} \quad \text{(Equivalent matching inductance, assuming } L_1' > 0\text{)} \tag{A.42}$$

$$\omega_{\text{tank}} = \frac{1}{\sqrt{L_2 C_3}} \quad \text{(Lower boundary of capacitive region)} \tag{A.43}$$

$$\omega_{\text{osc}} < \omega_{\text{tank}} \sqrt{1 + \frac{L_2}{L_1'}} \quad \text{(Upper boundary of capacitive region)} \tag{A.44}$$

$$L_1 \approx \frac{1}{\omega_{\text{osc}}^2 C_a} \quad \text{(Optimal matching inductor for antenna resonance)} \tag{A.45}$$

Appendix B

Scientific article describing this thesis

The thesis should also contain a short scientific article. If you write your thesis in Dutch, you must write the article in English, and vice versa. We advise you to employ the IEEE Manuscript Templates for Conference Proceedings (`https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn`). Compile the article in another project and include the generated pdf file as shown below:

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: “Wb/m^2” or “webers per square meter”, not “webers/m^2”. Spell out units when they appear in text: “. . . a few henries”, not “. . . a few H”.
- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not

Appendix C

Poster

Appendix D

GenAI code of conduct

Placeholder

GenAI code of conduct

Download the most recent KU Leuven GenAI code of conduct, fill it in, export the completed document to PDF, and manually include that PDF here.

The files `chapters/GenAI_Code of conduct_AJ 2526_*` are provided as a starting point only. Always check the KU Leuven website for the most recent version before submitting your thesis.

Replace this placeholder with, for example:

```
\includepdf [pages=-]{chapters/code-of-conduct.pdf}
```